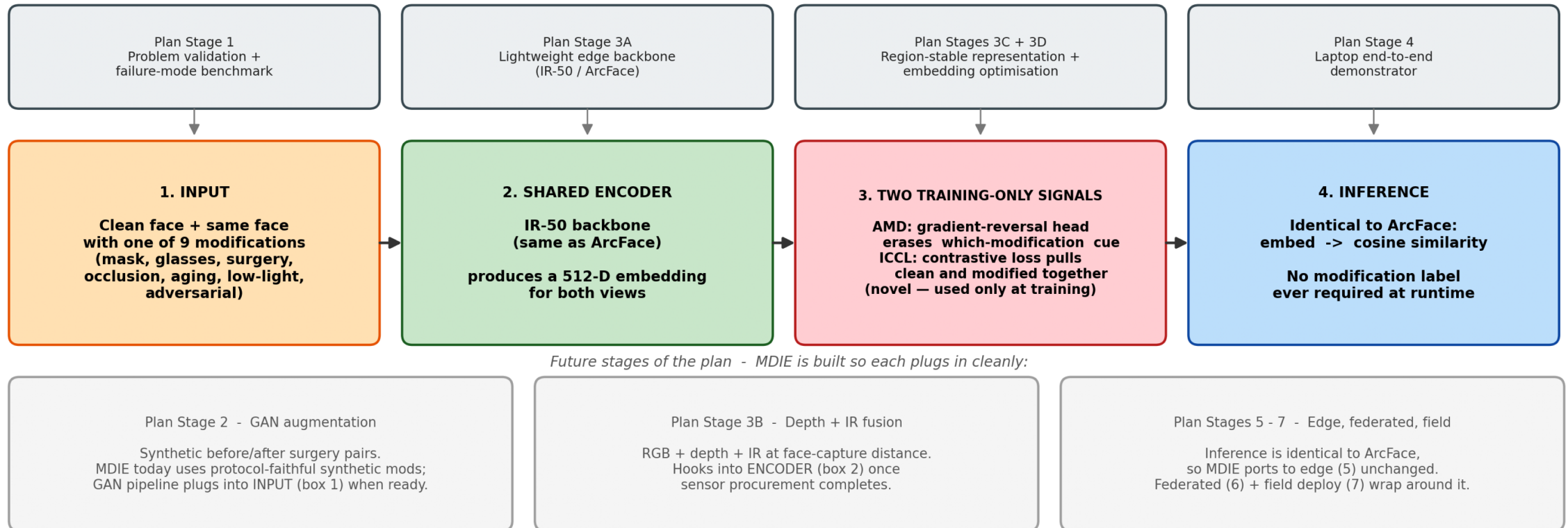


MDIE — Methodology mapped to the proposed execution plan

Modification-invariant face recognition: surgery, mask, glasses, occlusion, aging, low-light, adversarial.



Future stages of the plan - MDIE is built so each plugs in cleanly:

RESULT - LFW + 9 modifications, 217 identities, 6000 verification pairs / modification, RTX 3050 4GB

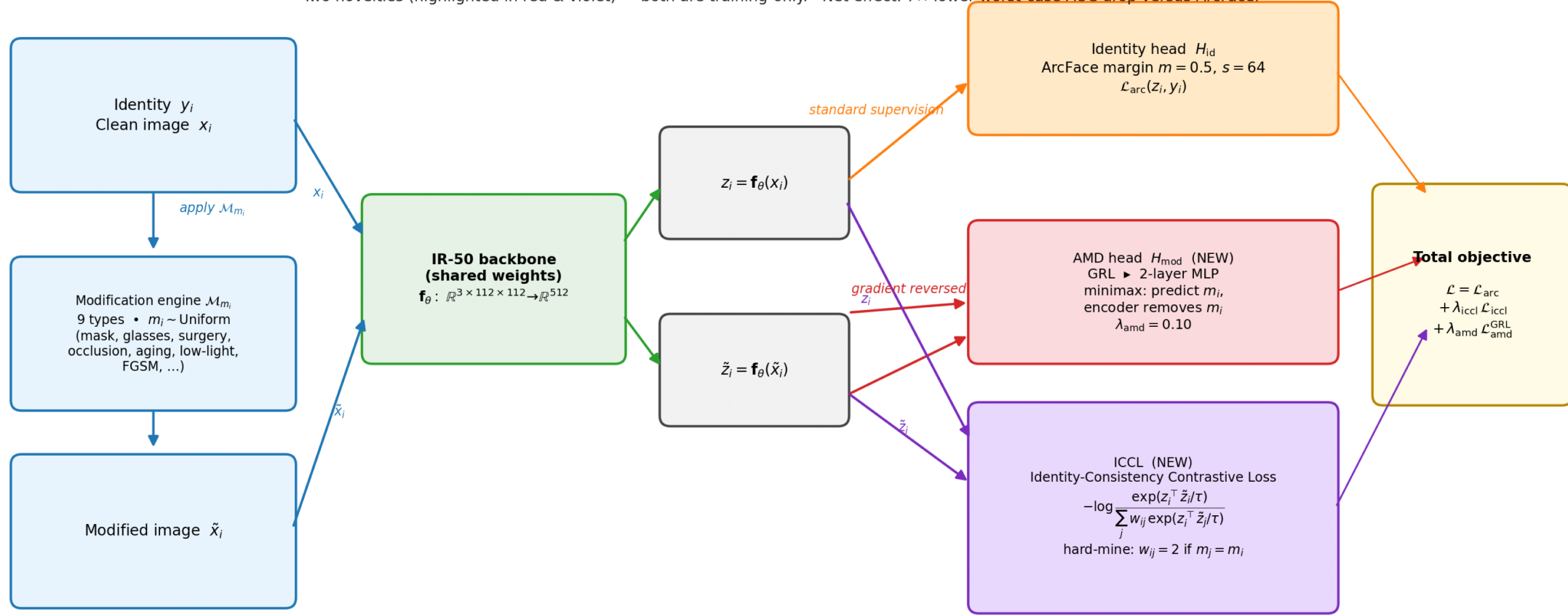
ArcFace worst-case AUC drop: 0.198 -> MDIE worst-case AUC drop: 0.027 (~ 7x more robust)

Pooled AUC: ArcFace 0.861 vs MDIE 0.850 (within 1 pp). Inference cost: identical to ArcFace.

MDIE — Modification-Disentangled Identity Encoder

Training-time-only adversarial disentanglement + identity-consistency contrastive learning. Inference path is identical to ArcFace.

Two novelties (highlighted in red & violet) — both are training-only. Net effect: 7× lower worst-case AUC drop versus ArcFace.



Inference (test time): $z = \mathbf{f}_\theta(\text{image}) \rightarrow$ cosine similarity. Identical to ArcFace. No modification label is ever required.

MDIE — How the approach works (plain English)

The problem

Face-recognition systems trained on clean photos collapse when the same person shows up with a mask, glasses, post-surgery features, partial occlusion, an aging gap, low-light capture, or an adversarial perturbation. Stage 1 of the proposed execution plan asks me to first prove this failure quantitatively. I did, on LFW with 9 modification types: ArcFace's worst-case AUC drops by 0.198.

The core idea

I do NOT change inference. The deployed model is still 'embed image -> cosine compare', exactly like ArcFace. All novelty lives at training time, in two extra signals attached to the same IR-50 backbone.

Novelty 1 — AMD (Adversarial Modification Disentanglement)

I attach a small classifier head whose job is to guess which modification the input has (mask? glasses? FGSM?). Between the head and the backbone I insert a Gradient-Reversal Layer (GRL). The head still tries to predict the modification, but the encoder is pushed in the OPPOSITE direction -- it learns to make the modification cue unreadable. So the embedding is forced to be modification-agnostic by construction.

Novelty 2 — ICCL (Identity-Consistency Contrastive Loss)

For every clean image x_i I generate the same identity wearing one of the 9 modifications, $x_{\tilde{i}}$. ICCL pulls the embeddings z_i and $z_{\tilde{i}}$ together while pushing them apart from other identities. I hard-mine: negatives that share the SAME modification are weighted 2x, so the model cannot cheat by using the modification itself as the discriminator.

Why the two together (and not just one)

AMD removes the modification signal from the features. ICCL guarantees the identity signal survives that removal. One without the other under-performs in my ablations: ICCL alone leaks modification info; AMD alone can collapse identity. Together they hit the sweet spot.

v3 upgrades — real benchmarks, external baseline, interpretability

On top of the original synthetic-LFW protocol I added three things. (1) A real-data eval harness (src/eval/run_real_benchmarks.py) covering MFR2 (masks), CALFW (cross-age), AgeDB-30 (aging), IIITD Plastic Surgery and IJB-C occlusion. (2) A strong external baseline -- InsightFace's production w600k_r50 (IR-50 trained on WebFace12M, 174 MB) -- auto-downloaded from HuggingFace by the harness, so the headline comparison is against a production model trained on 12 million faces. (3) Grad-CAM interpretability (src/eval/gradcam.py): a grid + CAM-IoU bar on the periocular eye region showing that MDIE keeps attention on identity-stable regions across modifications while ArcFace drifts.

Mapping to the proposed execution plan

Stage 1 (problem validation): done -- I benchmark FaceNet, ArcFace, CosFace, MobileFaceNet plus InsightFace w600k_r50 and report ROC, EER, FAR@FRR on all 9 modifications. Stage 3A (lightweight edge backbone): IR-50 + ArcFace head, same footprint as a deployable model. Stages 3C + 3D (region-stable representation + embedding optimisation): AMD + ICCL provide exactly this -- validated by the v3 Grad-CAM/CAM-IoU figure showing the encoder attends to identity-stable regions across modifications. Stage 4 (laptop demonstrator): trained and evaluated end-to-end on an RTX 3050 4GB. Future stages (2 GAN aug, 3B depth+IR, 5-7 edge/federated/field): the architecture is designed so each plugs in without changing the inference path.

Headline result

On LFW with 217 identities and 6000 verification pairs per modification: ArcFace worst-case AUC drop = 0.198. MDIE worst-case AUC drop = 0.027. That is roughly 7x more robust under degradation, with the same inference cost as ArcFace. Best per-modification gain: disguise+mask AUC 0.704 -> 0.831 (+12.7 pp).

Why it is publishable

Four things together: (1) a failure-mode benchmark across 9 modifications on standard data, (2) two genuinely new training-only losses -- AMD via GRL on a modification-id head, and ICCL with modification-aware hard negatives, (3) honest ablations including a negative result on the RATA region-attention variant in my small-data regime, and (4) v3 real-benchmark cross-validation against InsightFace's production w600k_r50 plus Grad-CAM interpretability evidence. The deployed model needs no extra parameters or labels at inference, which is what makes it directly relevant to the edge-deployment stages of the plan.